

Stephen King

Senior Python Developer & Software Engineer

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PROFESSIONAL SUMMARY

Over 18 years of experience with Python, with a practical and engineering-focused approach to software development. I prefer clear, well-structured code that is easy to test, maintain and extend, and I look for straightforward solutions without introducing unnecessary complexity. I look for opportunities to automate repetitive work and remove unnecessary manual steps. I take ownership of problems and am comfortable working independently, while also enjoying code reviews, technical discussions and working with other engineers. I keep my Python knowledge current through books and technical courses, and I track language updates directly through new PEPs. Newsletters like PyCoder's Weekly and Python Weekly help me stay on top of broader tooling and community news. I enjoy learning new technologies when they solve a real problem and am equally comfortable working with established codebases, modernising legacy software or building new systems.

CORE TECHNICAL SKILLS

Python Development	OOP & Design Patterns, SOLID Principles, Type Hints / typing, pytest, mypy, asyncio / async programming, Pydantic, REST APIs, Ruff, uv, Documentation, FastAPI, Django, DRF, Flask, TDD, modern Python Packaging, Performance Profiling (cProfile, py-spy, line_profiler)
Data & Systems	SQL, pandas, NumPy, Oracle, Polars, xarray, PostgreSQL, CouchDB, Elasticsearch, Redis, SQLAlchemy, Microservice Architecture, ETL
DevOps & Cloud	GitHub Actions, GitLab CI, Jenkins, Docker, AWS ECR, ECS, Fargate, Lambda, Terraform, Pulumi, AWS CDK, Ansible, boto3, Kubernetes, Infrastructure as Code (IaC)
Test Automation	pytest, Robot Framework, Test Framework Design, API Testing, Behave, Playwright, Selenium, CI/CD Test Integration, System / Integration testing
AI & Machine Learning (AI/ML)	LangChain, LangGraph, LlamaIndex, Claude Code, RAG Pipelines, Vector Databases, Agentic Workflows, MCP, PyTorch, Hugging Face Transformers, Prompt Engineering, Ollama, vLLM, Chroma, Qdrant, Sentence Transformers, Hybrid Search, Reranking, BM25

CAREER HISTORY

ToFollow | Senior Python Developer in AI | Oct. 2025 –

Designed and built production Retrieval Augmented Generation (RAG) platforms using Python, LangChain and LlamaIndex. Developed the retrieval pipeline from document ingestion and chunking through to embedding, vector search, reranking and LLM response generation, with source citations for retrieved content.

- Modular Platform Architecture:** Developed end-to-end RAG platforms using LangChain and LlamaIndex for application and pipeline orchestration. Built document ingestion, adaptive chunking, embedding generation, vector indexing, retrieval and response generation as separate components so individual stages could be tested and changed independently.
- Hybrid Vector Retrieval:** Combined dense vector embeddings with sparse keyword search in Qdrant to implement hybrid retrieval. Applied query-specific retrieval rules and metadata pre-filtering to reduce the search space before vector search and reranking.
- Structured Ingestion Pipelines:** Built ingestion pipelines for multiple unstructured document formats and normalised their content into a common representation for indexing and retrieval. The shared schema allowed new document types to be added without rewriting the retrieval pipeline.
- Asynchronous REST Services:** Built asynchronous REST APIs with FastAPI for RAG ingestion, retrieval and generation. Added document management, health and status endpoints, with support for concurrent requests and streaming responses. Kept retrieval, ranking and generation as separate pipeline stages.
- Embedding & Reranker Evaluation:** Evaluated fine-tuned embedding models and cross-encoder rerankers using Hugging Face Transformers and Sentence Transformers. Measured Mean Reciprocal Rank (MRR) against technical documentation to assess retrieval performance and compare reranking approaches.

- **Empirical Evaluation Pipelines:** Built RAG evaluation pipelines with ragas to measure context precision and answer relevance across different index configurations. Used measured results to select chunking strategies rather than relying on fixed configuration.
- **Interactive Debugging Interfaces:** Built Streamlit and Dash interfaces for testing and debugging RAG pipelines. Added live response streaming and chunk-level inspection so retrieved text and source documents could be examined during development.
- **Automated CI/CD Delivery:** Configured GitHub Actions CI/CD pipelines to run automated tests, build and publish Docker images to AWS ECR, and deploy services to AWS ECS on EC2. Automated the release process so deployments were repeatable with minimal manual intervention.

Tech: Python, Claude Code, LangChain, LlamaIndex, FastAPI, Qdrant, Pinecone, Chroma, Hugging Face Transformers, Docker, AWS ECR, ECS, GitHub Actions, Git, Streamlit, Dash, ragas, Guardrails, metadata filtering, Jira, Confluence

Met Office | Senior Scientific Software Engineer (Python) | Dec. 2024 – Jul. 2025

Replaced legacy Fortran MORST post-processing pipelines with Python and Cylc workflows centered on the JULES (Joint UK Land Environment Simulator) surface exchange model. The replacement processing combined STF JULES output with StaGE meteorological data and produced data in the formats required by downstream systems.

- **Forecast Pipeline Modernisation:** Developed Python and Cylc workflows for the next-generation surface transport forecasting system, using STF JULES to process output from the UKV, MOGREPS-UK and Global NWP weather models.
- **Fortran Migration:** Migrated three legacy Fortran datafeeds to Python, removing three dependencies on legacy code and reducing the amount of Fortran that needed to be maintained.
- **Cylc Workflow Orchestration:** Designed and built workflow orchestration with the Cylc engine, developing Directed Acyclic and Cyclic Graph (DAG/DCG) workflows for complex meteorological post-processing. Automated extraction and transformation of data from the MOGREPS UK, UKV and Global weather models.
- **Meteorological Data Processing:** Processed and transformed high resolution, multidimensional meteorological datasets in NetCDF format, working with xarray, NumPy, pandas, and SciPy. Used Jupyter notebooks to develop and validate data transformation and processing logic. Downstream scientific applications received standardised datasets.
- **Environment Management:** Configured Conda and Mamba environments while integrating JFrog Artifactory for internal package distribution. Version control of dependencies ensured reproducible development setups across the team.
- **Unit Testing & Validation:** Developed test suites in pytest using monkeypatching and parameterised fixtures to isolate external dependencies. Test coverage reached over 99%, establishing a verification standard for ongoing development.
- **Automated Release Pipelines:** Built CI/CD pipelines with GitHub Actions to run test suites and manage deployments for the post-processing platform across AWS and Azure environments. Automated validation eliminated manual release procedures.

Tech: Python, Fortran, Cylc, MASS, MOOSE, NetCDF, xarray, NumPy, pandas, SciPy, Conda, Mamba, JFrog Artifactory, Git, GitHub Actions, AWS, Azure, Jinja2, Jira, YAML, TOML, Linux, Shell/Bash scripts

Freelance | Senior Software Engineer / Python Developer | Sep. 2023 – Nov. 2024

Delivered independent software development for IoT, medical diagnostics, and robotics clients, including a fully automated CI/CD release process and a computer vision diagnostics pipeline for clinical use.

- **Client Engineering Delivery:** Delivered software engineering services for startup IoT platforms, medical diagnostic devices, and robotic SDKs. Projects shipped production software tailored to client specifications on predictable schedules.
- **Automated Release Pipelines:** Built CI/CD workflows using GitHub Actions to run test suites and package Python applications. Configured Ansible playbooks to publish build artifacts to AWS S3 and automate deployments to AWS EC2 instances.
- **GitOps Deployment Infrastructure:** Implemented declarative application deployment workflows using ArgoCD for continuous delivery. Version-controlled Git repository states drove environment synchronisation and provided clear release audit trails.
- **Computer Vision Diagnostics:** Developed image processing and object detection pipelines using OpenCV alongside PyTorch to automate bacterial colony identification. Applied Hugging Face transfer learning models and YOLOv8 object detection to score antibiotic efficacy without requiring manual image analysis.

- **Template & Process Tooling:** Authored Jinja2 configuration templates for deployment scripts and managed client project backlogs within Jira alongside Confluence. Standardised project documentation kept team members aligned during active development cycles.

Tech: Python, GitHub Actions, AWS S3, AWS EC2, Ansible, ArgoCD, OpenCV, PyTorch, Hugging Face, YOLOv8, Jinja2, Jira, Confluence, IoT, Medical Diagnostic Devices, Robotic SDKs, Computer Vision.

AMD | Senior Python Developer | Jan. 2020 – Jul. 2023

Led the design of an automated CI testing framework and network automation tooling across a large test estate. The resulting architecture improved deployment reliability while accelerating test execution cycles.

- **CI Framework Architecture:** Led an engineering team to design and build an automated CI testing framework, exposing execution controls through a FastAPI service hosted on AWS with MongoDB as the data store. Event-driven orchestration using AWS Lambda with boto3 made the automation scalable and reduced operational overhead.
- **Network Infrastructure Automation:** Automated network device provisioning across the test estate using Nornir and Netmiko alongside NAPALM, Fabric, Ansible playbooks, and Pulumi for Infrastructure as Code. Standardised configuration scripts eliminated manual setup errors and sped up test topology deployments.
- **Data Layer Integration:** Integrated relational and NoSQL data sources using PostgreSQL and SQLAlchemy ORM for backend service persistence. The unified data access layer simplified service integration while maintaining application uptime during backend scaling.
- **Web & Load Validation:** Developed functional test suites and performance benchmarks for web interfaces using Playwright alongside Locust. Early execution of automated scenarios caught functional bugs and performance bottlenecks before production release.
- **Workflow Gap Analysis:** Identified execution bottlenecks in existing development workflows and introduced targeted automation tools. Eliminating redundant steps increased test suite reliability and reduced overall build turnaround times.
- **Agile Engineering Execution:** Participated in Agile ceremonies including sprint planning, backlog refinement, and daily standups. Prioritised competing work to meet sprint deadlines, shipping high priority features consistently on schedule.
- **Code Quality & Mentorship:** Conducted peer code reviews and mentored junior developers on Python best practices. Shared engineering standards reinforced software quality while accelerating team onboarding.

Tech: Python, FastAPI, MongoDB, AWS, AWS Lambda, boto3, Nornir, Netmiko, NAPALM, Fabric, Ansible, Pulumi, Infrastructure as Code (IaC), PostgreSQL, SQLAlchemy, ORM, Playwright, Locust, Agile, Git.

Career Break | Jan. 2019 – Nov. 2019

- I took this period to renovate the house and relocate.

Qualcomm | Automation Engineer | Aug. 2016 – Dec. 2018

Maintained and extended an in-house Python test automation framework while leading a code migration from Python 2 to Python 3. The updated codebase maintained zero regressions and reached 100% unit test coverage across the suite.

- **Framework Development:** Maintained and expanded the internal Python test framework by fixing defects and adding features for CI/CD pipelines. Standardised test routines cut test failure rates and sped up build verification.
- **Python 2 to 3 Migration:** Led the codebase migration from Python 2 to Python 3 across the entire automation suite. Code updates completed with zero functional regressions and simplified long-term maintenance.
- **Desktop Utility Tooling:** Built desktop graphical utilities using PyQt5 to assist engineering teams during test runs and debugging sessions. Interactive interfaces reduced manual diagnostic time during test failures.
- **Unit Testing Coverage:** Authored unit test suites using pytest to reach 100% test coverage across core modules. Full test coverage verified framework logic during the Python 3 upgrade.
- **Release & SCM Integration:** Extended automation tooling for Software Configuration Management alongside code review workflows in Git and Gerrit. Automated checks enforced release consistency and removed manual execution steps.
- **Developer Mentoring:** Conducted technical training sessions and mentored engineers on Python development best practices. Broader team adoption reduced onboarding time for engineers writing automated test suites.
- **Toolchain Consolidation:** Consolidated standalone testing tools into a unified automation workflow. Eliminating redundant tool steps streamlined daily engineering operations.

Tech: Python, PyQt5, pytest, Git, Gerrit, Jenkins, CI/CD, Bash, Linux, REST APIs, SCM

Cisco | *Python Developer in Test (Devops)* | Jul. 2011 – Aug. 2016

Developed automated tests and test infrastructure for ScanSafe, Cisco's cloud-based web security platform, covering web traffic scanning, authentication, configuration, distributed services, performance and system integration across multiple platforms and environments.

- **Web Test Automation:** Created automated UI and functional test suites using Python alongside Java with Selenium, covering ScanSafe functionality across multiple platforms, environments and customer configurations.
- **Framework Architecture:** Built reusable test frameworks and infrastructure that supported functional, regression and integration testing throughout the software development lifecycle. Standardised test components cut maintenance work and allowed engineering teams to share test scripts across products.
- **CI/CD Integration:** Integrated automated test suites into Jenkins pipelines. Builds ran verification checks on every commit, giving developers immediate feedback on build health.
- **Infrastructure Automation:** Automated deployment and configuration of test environments with Ansible playbooks and Puppet manifests, reducing manual setup and keeping development and test systems consistent.
- **Microservices Validation:** Designed and executed automated system integration tests on microservice architectures backed by CouchDB and Memcached. Verified message queue performance using ActiveMQ alongside Redis, and verified web caching behavior through Squid proxies.
- **Environment Virtualisation:** Configured virtualised test environments using VMware and VirtualBox, while setting up KVM hypervisors on Linux test nodes. Later adopted Docker containers to give developers identical local and CI setups.
- **Load & Analytics Testing:** Executed load and performance test scenarios using Apache JMeter. Benchmark runs measured throughput and query execution times on Vertica analytics databases, uncovering database bottlenecks before production deployment.
- **System Metrics Telemetry:** Set up system monitoring and log aggregation using the ELK Stack alongside Grafana dashboards. Tracked real-time metrics with Graphite and Prometheus to identify resource spikes during extended stress tests.
- **Network Traffic Analysis:** Diagnosed low-level network issues using Wireshark and tcpdump. Scanned target nodes with nmap to verify port configurations and locate packet drops across distributed network services.
- **Took part in code reviews and Agile ceremonies** to improve sprint productivity. Used Jira to track defects and development work within an Agile development process.

Tech: Python, Java, Selenium, Jenkins, Ansible, Puppet, CouchDB, Memcached, ActiveMQ, Redis, Squid, VMware, VirtualBox, KVM, Docker, SAML, SSO, OAuth 2, Ping Identity, OpenAM, MS ADFS, JMeter, Vertica, ELK Stack, Grafana, Graphite, Prometheus, Wireshark, tcpdump, nmap, Jira, Agile, Linux RHEL.

Solarflare Communications | *Python Developer in Test* | Mar. 2010 – Jun. 2011

Built Python-based test automation for 10Gb Ethernet NICs and switches, covering network protocol compliance, performance testing and competitive benchmarking.

- **Automated Test Framework:** Built a Python test framework that brought together a large suite of automated scripts with open-source and commercial test tools to test 10Gb Ethernet NICs and switches. The framework covered both protocol functionality and device performance across multiple hardware platforms.
- **NIC and Switch Validation:** Validated 10Gb network interface cards and Ethernet switches under line-rate traffic conditions. The test suite identified packet loss and protocol edge-case errors before hardware reached production.
- **Performance Benchmarking:** Measured latency and throughput against competing network devices, creating visual charts with Matplotlib. Engineering teams relied on these charts to adjust hardware configurations and tune driver settings.
- **Protocol Compliance Automation:** Created protocol test specifications from IETF RFCs and IEEE standards, then automated compliance testing with IXIA packet generators and Python Scapy. This gave the team repeatable tests for network protocol behaviour and reduced manual protocol testing.
- **Test Result Dashboards:** Built web reporting dashboards with Django to aggregate execution logs and performance metrics. Clear visibility into historical test runs allowed hardware engineers to spot regressions early.
- **Quality Process Specifications:** Worked with engineering and product teams to define test strategies, test specifications and quality processes for 10Gb Ethernet products, covering both protocol compliance and performance validation.

Tech: Python, Ixia Packet Generators, Scapy, Matplotlib, Django, 10Gb Ethernet, 10Gb NICs, Network Switches, TCP/IP, UDP, IETF RFCs, IEEE Standards, Linux.

ADDITIONAL EXPERIENCE

- Citrix, QA Team Leader
- BT, Network Support Analyst: 3rd line support of ISP Network infrastructure.
- 3Com, QA Analyst: Network protocol verification in switches and routers.
- Nokia, QA Analyst

SECURITY CLEARANCE

- SC (Security Check): lapsed July 2026, previously held, eligible for expedited re clearance.
- DBS (Disclosure and Barring check): Current.

EDUCATION

- MSc. Biochemistry, University of Reading.
- BSc. Biochemistry with Chemistry, University of Reading.
- 'A' Levels: Physics, Chemistry, Pure Mathematics and Biology.

PROFESSIONAL CERTIFICATIONS

- AWS Certified Cloud Practitioner (expired).
- Cisco: CCNA, CCNP, CCDA, CCDP (obtained ~2008–2010, not actively renewed).
- Citrix: CCA, CCEA (obtained ~2006–2008, not actively renewed).